Object Oriented Programming :: Relationships

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OOP

The Four Pillars
OOP
Abstraction
Inheritance
Polymorphism
Encapsulation

OOP :: Class/Object/Instance

What is Class ???

- A Template or Blueprint
- Contains collection of properties & methods
- Helps to build Objects with different characteristics & behaviors

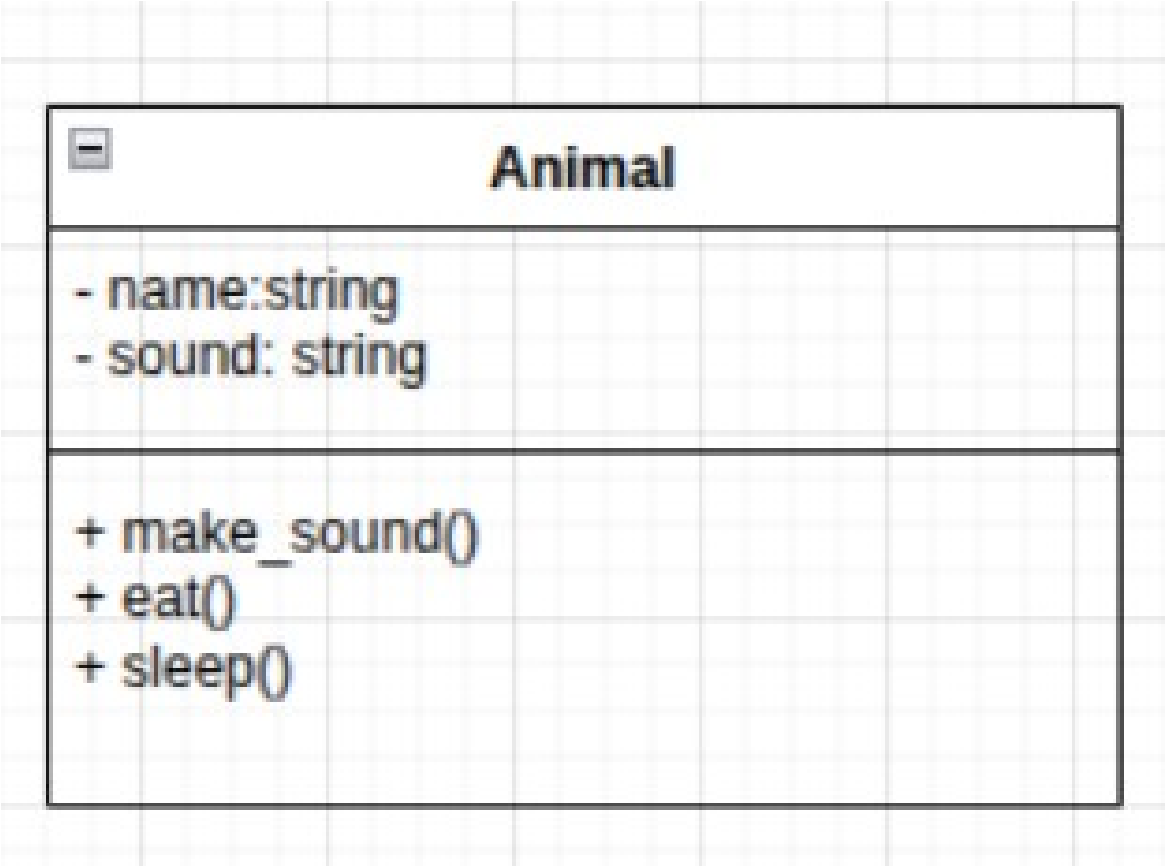
What is Object ??

- Entity that we implement in SW
- Also called "**Object of Class**"
- Used to create Instances

What is Instance ???

- Virtual copy of the Object
- A given/achieved object is **Instance**
- When create **Instance** of class, distinct **Objects** will be created in Memory.

Object Oriented Programming :: Class



HAS
TYPE - > Animal

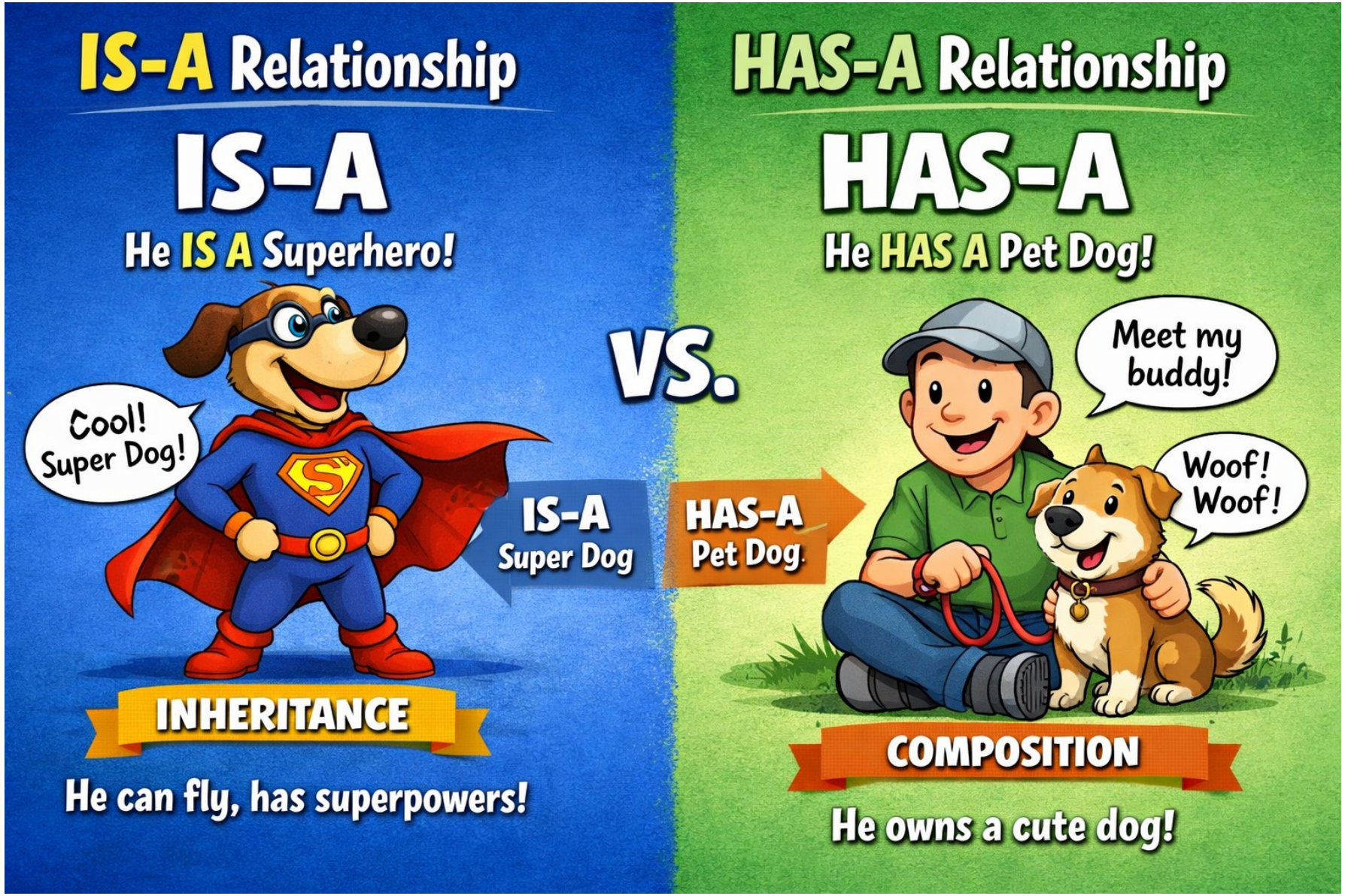
HAS Private variables " - "
HAS Public variables " + "

HAS Private methods " - "
HAS Public methods " + "

** Variable ??
Can hold some value

** Property ??
An attribute of a variable

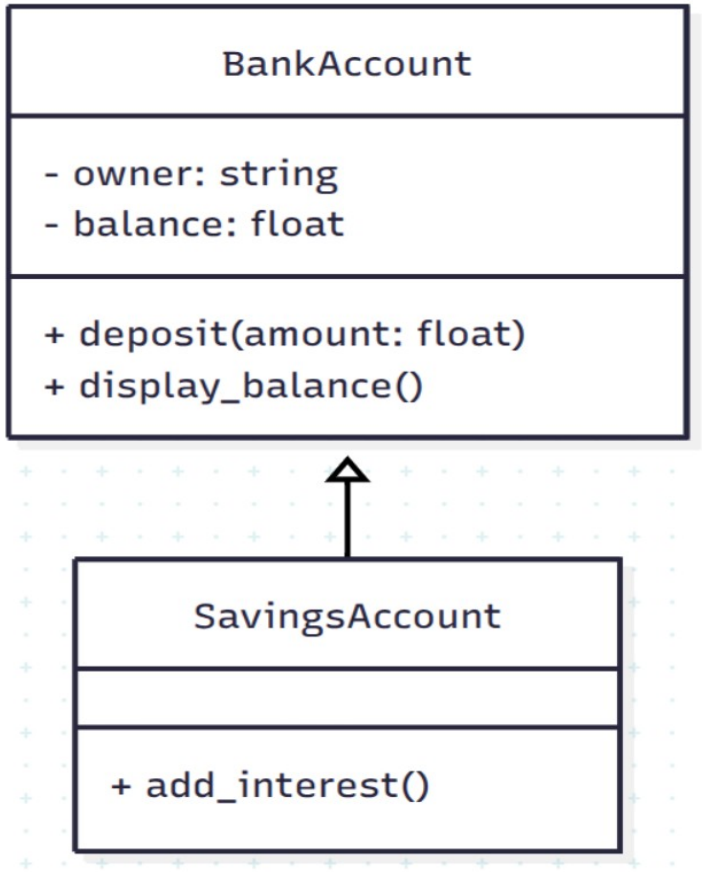
Object Oriented Programming :: Composition



Object Oriented Programming :: Inheritance/Composition

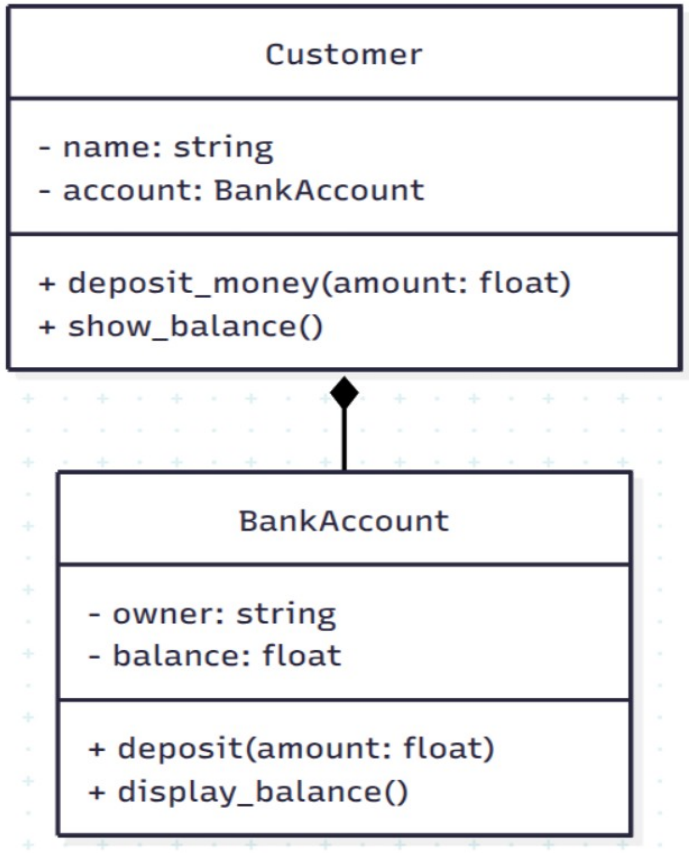
IS A

- Child extends parent class
- Tight coupling



HAS A

- Class contains anther class
- Loose coupling



SavingsAccount is a BankAccount → Inheritance

Customer has a BankAccount → Composition

Object Oriented Programming :: Association/Aggregation

Object Oriented Programming :: Relationships

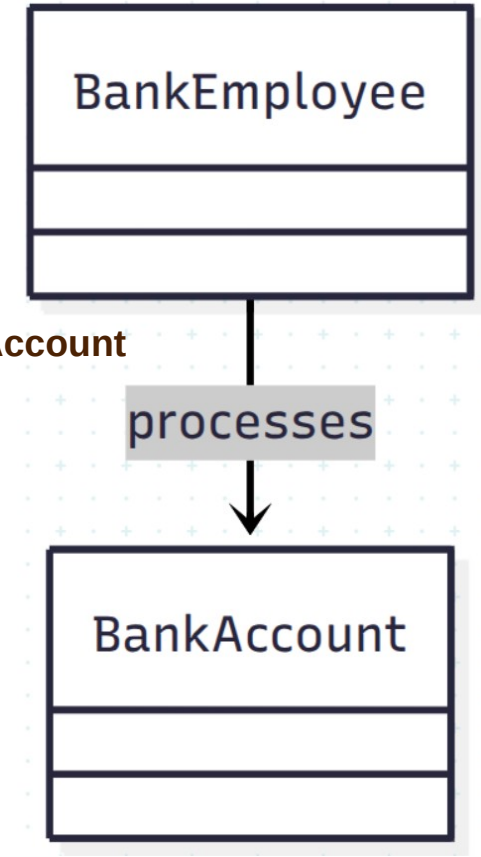
Association

- No strong ownership
- Objects can exist independently
- Just interaction

✔ BankEmployee does NOT own BankAccount

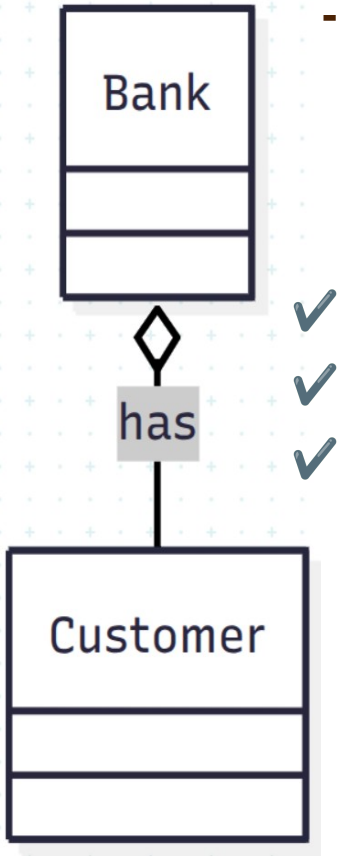
✔ It just interacts with it

✔ Both objects can exist independently



Aggregation

- Bank has Customers
- If bank closes, customers can exists



✔ Customer objects are created outside Bank

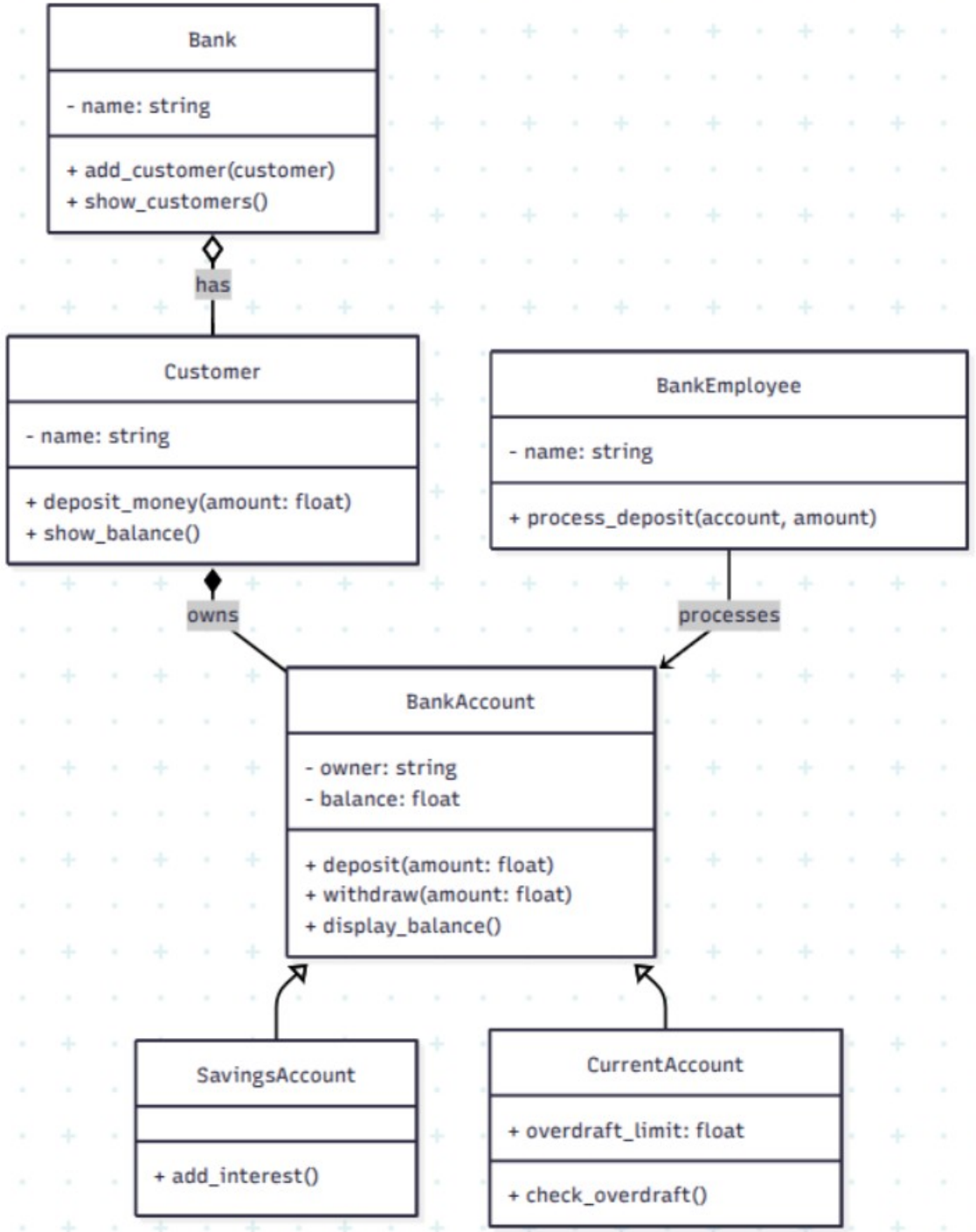
✔ Bank only stores references

✔ If Bank object is deleted → Customers still exist

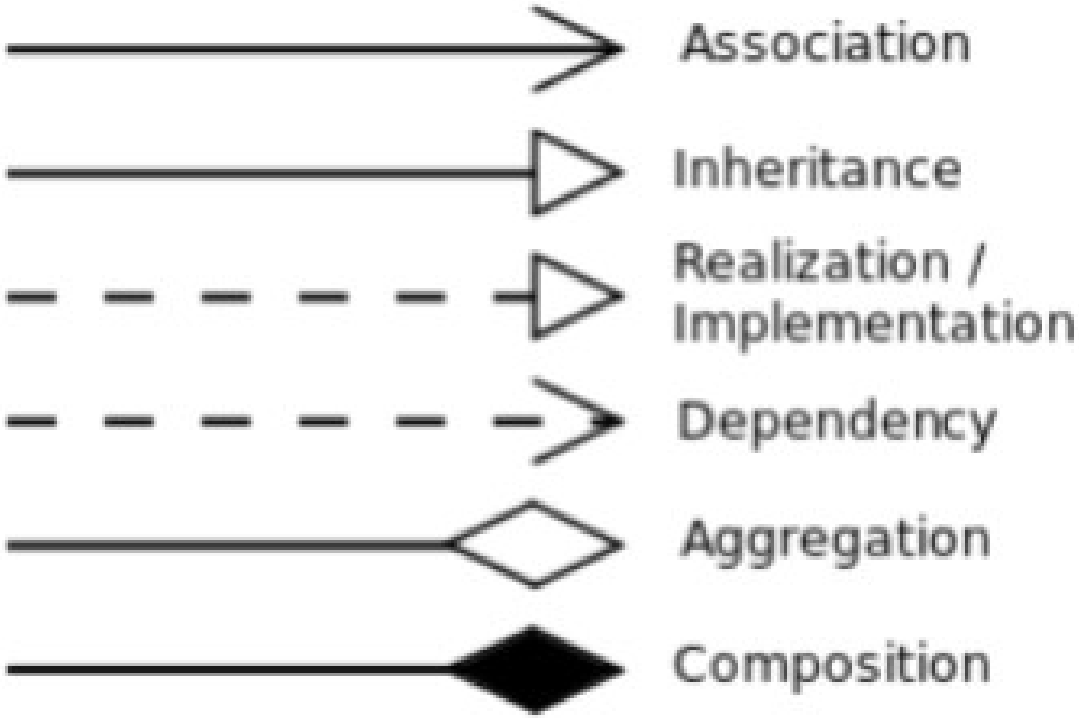
Object Oriented Programming :: Relationships

Concept	Meaning	Ownership	Example
Association	Uses–A	No ownership	Employee uses Account
Aggregation	Has–A (weak)	Weak ownership	Bank has Customers
Composition	Has–A (strong!)	Strong ownership	Customer has Account
Inheritance	IS–A	Parent-child	SavingsAccount is BankAccount

Relationships



OIP Classes :: Relationships



Association → “We know each other” 🤝

Inheritance → “Like father, like son” 👨➡️👦

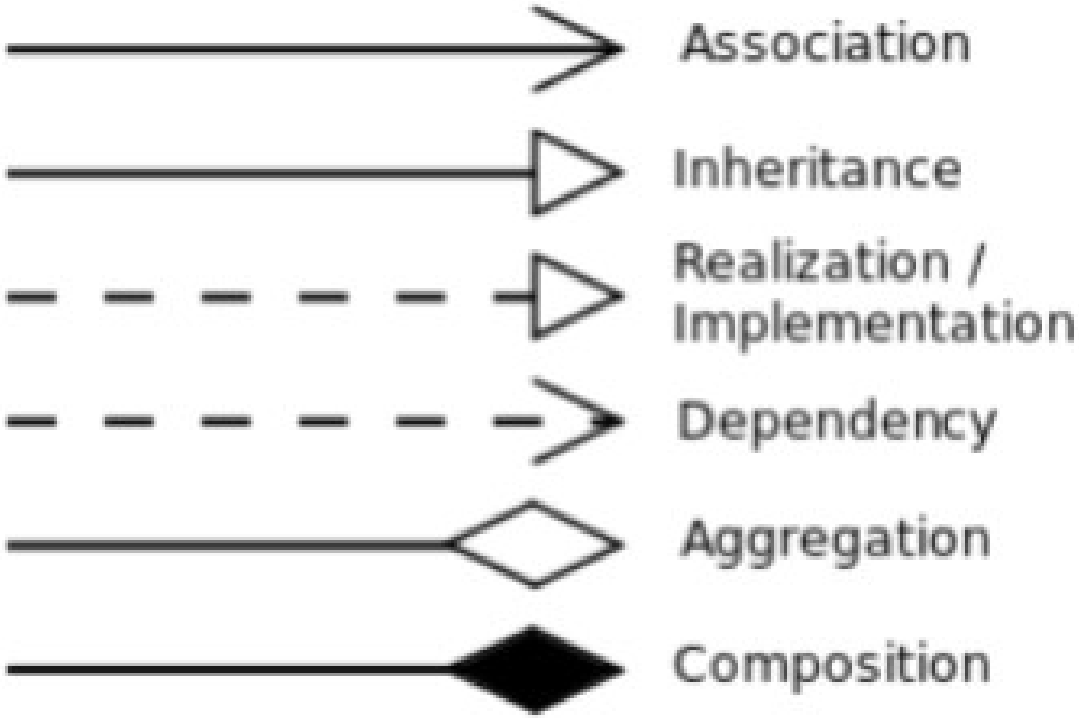
Realization → “Promise kept!” ✔️

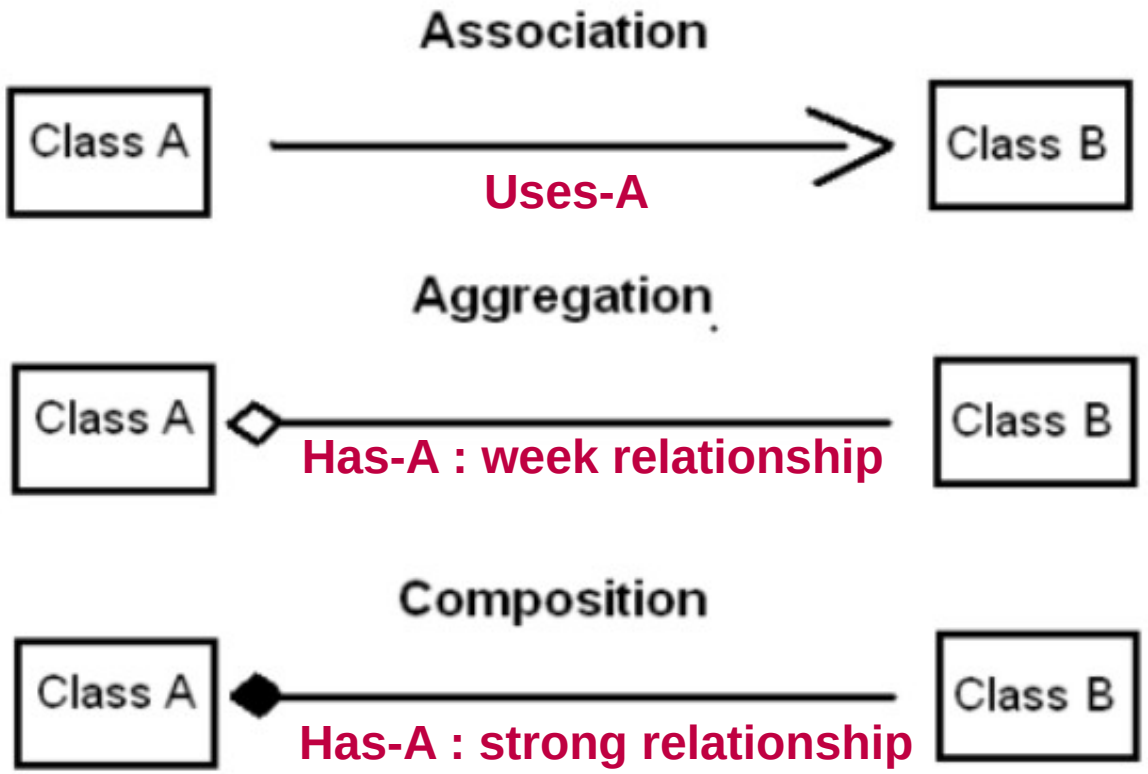
Dependency → “I need you sometimes” ☕

Aggregation → “We hang out together” 🎒

Composition → “I can’t live without you” ❤️

OIP Classes :: Relationships





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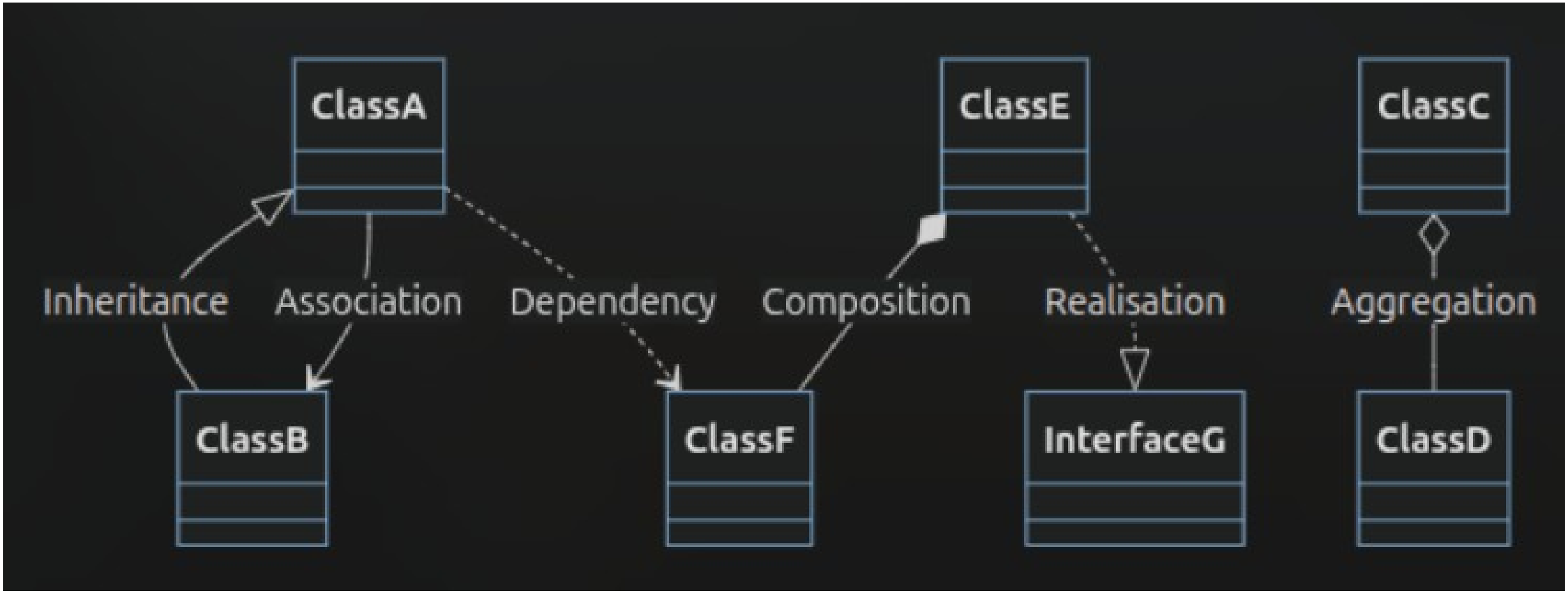

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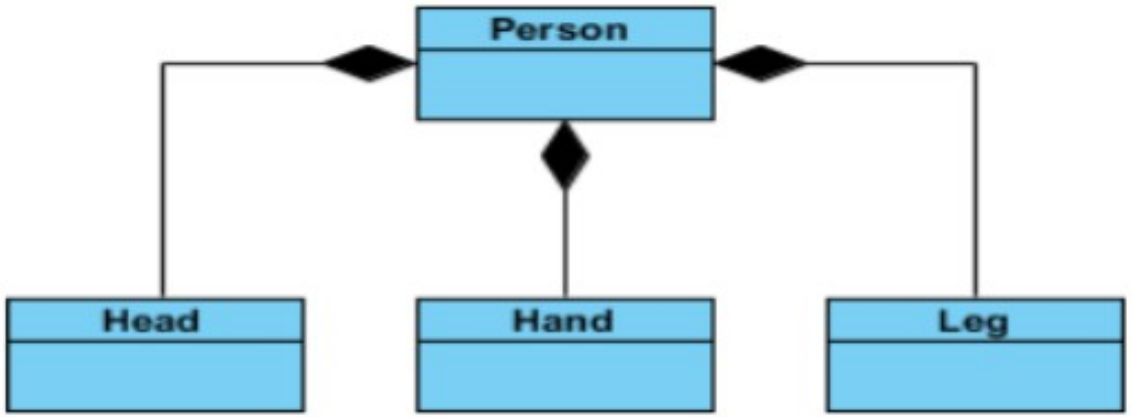

Composition: every car has an engine.

Aggregation: cars may have passengers, they come and go